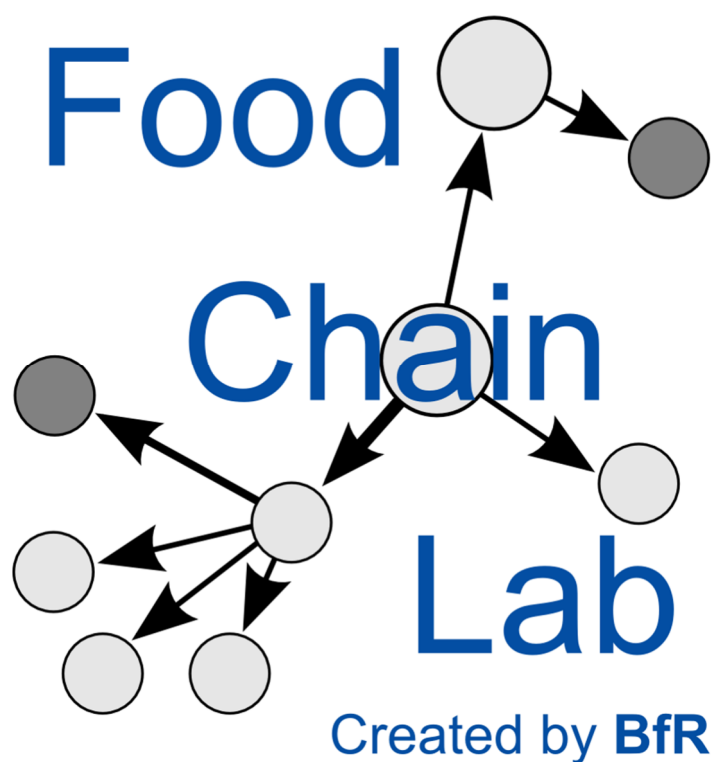




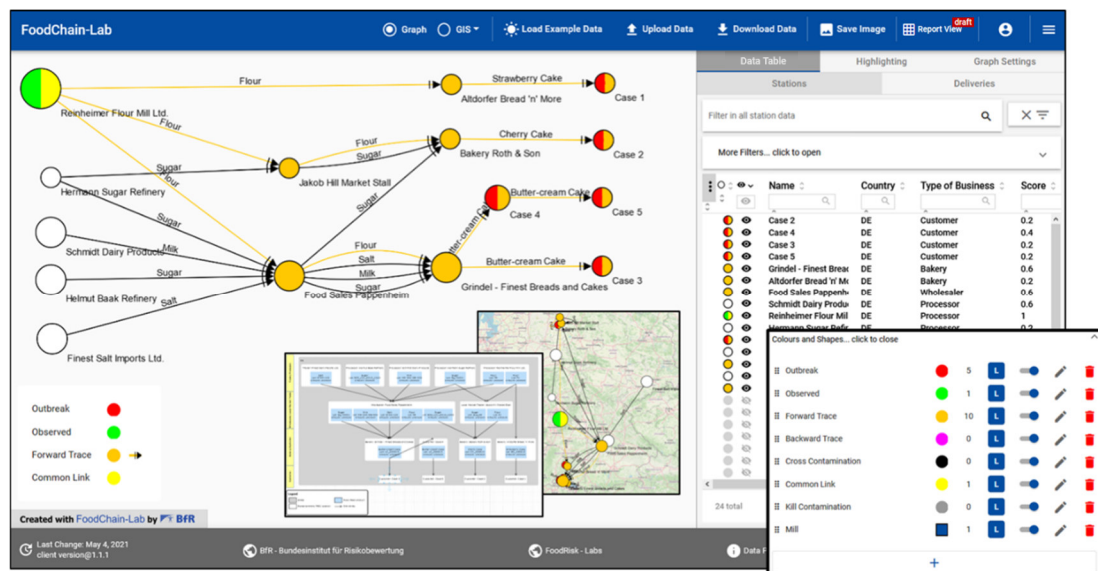
The FoodChain-Lab web application

FCL Web received funding from the EFSA-BfR Framework Partnership Agreement (FPA) GP/EFSA/AMU/2016/01 and GP/EFSA/AMU/2020/02 as well as the European Union's Horizon 2020 research and innovation programme under Grant Agreement No 773830 (One Health European Joint Programme, COHESIVE) and was supported by budget funds of the BfR.

The FoodChain-Lab web application: An innovative tool for supply chain tracing during foodborne incident investigations

The FoodChain-Lab web application (FCL Web) is free and open-source software and allows for fast and reliable tracing of suspicious food items along complex global supply chains during foodborne incidents. It unifies supply chain tracing data visualisation, analysis and reporting in one intuitive modular framework. Although running in a browser, data remains completely on the client side to ensure data protection. FCL Web is a joint output of One Health EJP COHESIVE and projects of the EFSA-BfR Framework Partnership Agreement and is used by U.S. FDA and several EU Member States.

Learn more about FoodChain-Lab at <https://foodrisklabs.bfr.bund.de/foodchain-lab/>.



FCL Web visualising the supply chain network of a fictitious foodborne disease outbreak in the network view, the map view, the report view and in the data table.

FCL Web is available at <https://fcl-portal.bfr.berlin>. All interested users can register and use FCL Web.

The software code is published at <https://github.com/SiLeBAT/fcl-client> and <https://github.com/SiLeBAT/fcl-server> under the GNU General Public License v3.0.

Quick Guide

Access to FCL Web

Please go to the FoodChain-Lab web application (<https://fcl-portal.bfr.berlin>). It is recommended to use a desktop PC or laptop with Chrome or Firefox as browser for the optimal user experience. Using a mobile phone or tablet is not recommended, as they may not provide optimal functionality and visibility.

Registration:

If you are a new user, registration is required to access FCL Web. Follow these steps to create an account:

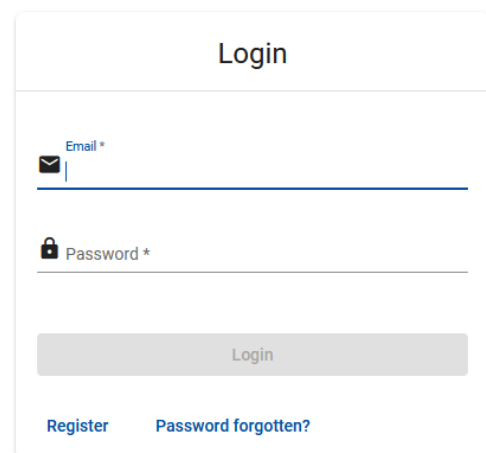
1. Visit the registration page at <https://fcl-portal.bfr.berlin/users/register>.
2. Fill in your details, including your email address and a strong password.
3. Confirm your email address by clicking the link sent to your inbox.
4. Your account will be activated by the FCL administrator. You will receive a notification via email once the activation is complete.
5. Log in with your registered credentials.

Login:

For returning users, login is straightforward:

1. Go to the FCL Web login page.
2. Enter your registered email address and password.
3. Click the **Login** button to access the application.

If you encounter issues logging in, ensure your credentials are correct. Use the **Password forgotten** link if you need to reset your password.



The screenshot shows a web form titled "Login". It features two input fields: "Email *" with an email icon and "Password *" with a lock icon. Below these fields is a "Login" button. At the bottom of the form, there are two links: "Register" and "Password forgotten?".

Troubleshooting

If you experience problems with FCL Web (e.g. it is not responding), it might help to

- try a different browser (Chrome or Firefox recommended)
- go to the dashboard ("FoodChain-Lab" button on the left side of the blue menu bar) and then click on "Tracing View" button
- reload the webpage via the browser refresh button, click on "Tracing View" button (if needed) and load the dataset

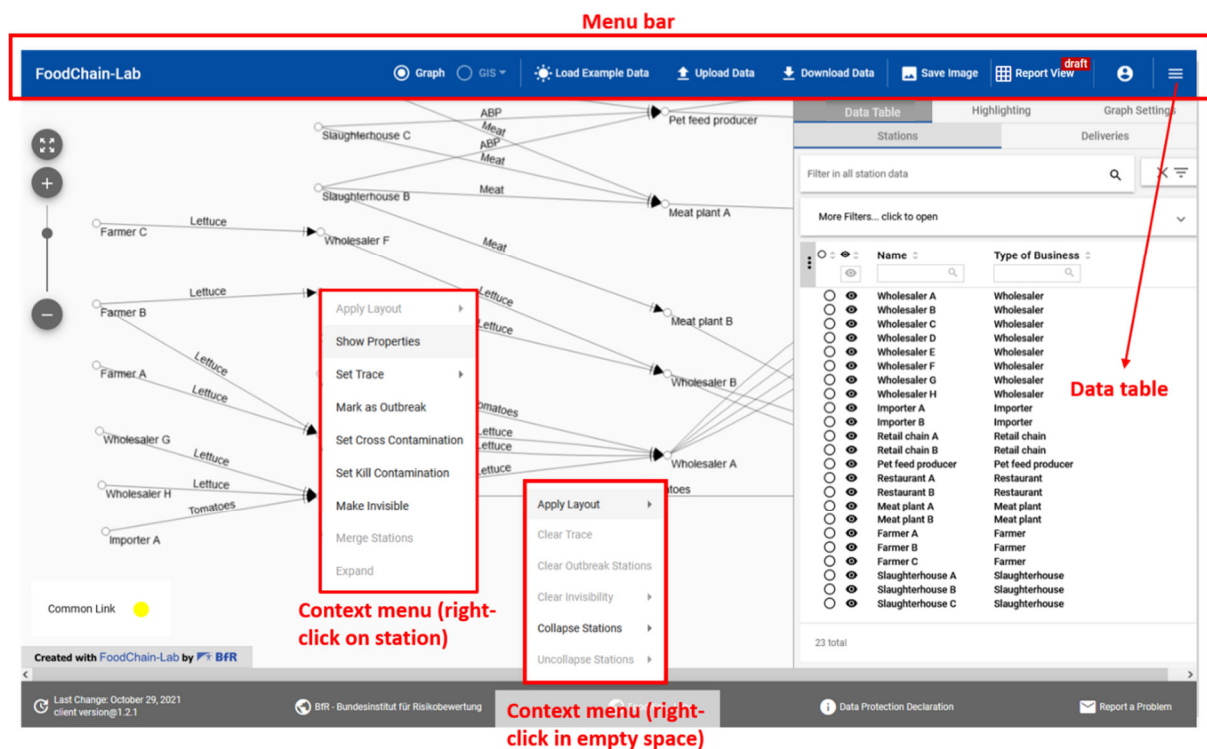
General information

Most functions in FCL Web can be accessed

... either by right-click on stations/deliveries/empty space

... or in the blue menu bar on the header of the app

The following screenshot gives you a quick overview on FCL Web.



Main functionalities in the spotlight

Disclaimer:

The scenarios and all companies as well as products are fictitious. If there is by chance any similarity to existing food businesses, products or outbreak investigations, this was not intended.

Left-click on the button “[Load Example Data](#)” and select “Example Data”.

You can see stations (circles) that represent various types of food business operators. The arrows are deliveries. The text in the middle of the arrows is the product which is delivered from one station to the other.

Nice to know: You can also upload your own tracing dataset by clicking the “Upload Data” button in the menu bar and upload your data in the FCL JSON format or the FCL All-in-One Excel template.

Apply the farm-to-fork layout (or a different layout) by right-clicking into the empty space -> Apply Layout -> Farm-to-fork.

If needed, make further adjustments to the network by moving the stations via drag-and-drop. The selection of multiple stations is possible. Nice to know: If you need to adjust the node size or font size, open the graph settings by clicking on the menu (three stripes) in the right upper corner, click on the “Graph Settings” tab. You can close the tab by again clicking on the menu button.

To indicate clusters of disease cases, right-click on the respective station (Case 1 to 5) and select “Mark as Outbreak” from the context menu. The sizes of the stations will change due to the changed scores. Have a look at the scores: Right-click on a station and click on “Show properties”). A common link is highlighted in yellow. It has a score of 1 and links all the disease cases.

Nice to know: The score is a measure for the number of disease cases to which each company and delivery is connected via deliveries. The higher the score, the more connections to cases. This helps you to prioritize the next steps of your tracing investigation since companies with higher scores might be more important in the situation. However, you need more evidence in addition to the tracing investigation to evaluate the situation.

To show the forward and backward trace of a specific station and the flow of products you can right-click on the station and select “Set Trace” and “Full trace”.

To reset the trace, right-click into the empty space and select “Clear Trace”.

You might want to create a picture of the supply chain network for a situation report (e.g. a Rapid Outbreak Assessment or other reports). Click on the “Report View” button in the upper right corner and click “Generate Report”.

Nice to know: You can customize the variables and the standard text which should be used to create the Report View in the Report Configuration dialogue before clicking “Generate Report”.

You can customize the Report View, e.g. by adding additional text fields or arrows. Drag-and-drop a yellow box from menu on the left and write some additional information into it (e.g. the number of cases at each restaurant).

Go to “File” -> “Print...” in the menu and check “One page” to save the Report View as PDF.

Go to “File” -> “Save” to save the Report View as XML. You can reload this file in the Report View of FCL Web at any time to make further adaptations.

Click on the “Tracing View” button in the upper right corner and “OK” to get back to your graph view.

Nice to know: As the Report View is only for reporting purposes, changes made in the Report View do not feed back to the Tracing View.

To use the data table and its filtering function, click on the menu button in the upper right corner (three stripes).

- Select the “Filter” tab and then the “Stations” or the “Deliveries” sub-tab.
- Here, all information on companies/deliveries is listed.
- Click on the three dots  in the table header to add further columns to the table.

Nice to know: There are more filter options available above the table. You can reset all filters by clicking the  button.

The Highlighting tab allows you to change or add labels for stations and deliveries as well as to set station shapes and colours for stations and deliveries in the respective sub-tabs for stations and deliveries.

In the “Stations” sub-tab in the “Label” function you can also anonymise company names using automatised anonymisation presets which you can customise according to your needs and the available data in your dataset (e.g. data on type of business, country, ...). Product names could be anonymised by using product categories as a “Label”, for example, if available in your dataset, or by switching off delivery labels.

Save an image of the graph via the “Save Image” button in the blue menu bar on the top.

Save the analysis using the button “Download Data” from the blue menu bar. You could reload this JSON file in the Tracing View module at any time to make further adaptations.

Further information

Do you need training in FCL Web?

Please do not hesitate to contact foodrisklabs@bfr.bund.de for online or on-site training opportunities.

Your feedback is essential:

Which functionalities do you miss? Do you have ideas on how to improve the existing software features? Please provide feedback to foodrisklabs@bfr.bund.de.

Contact

BfR traceability team
foodrisklabs@bfr.bund.de

About the BfR

The German Federal Institute for Risk Assessment (BfR) is a scientifically independent institution within the portfolio of the German Federal Ministry of Food and Agriculture (BMEL). It advises the Federal Government and the federal states (“Laender”) on questions of food, chemicals and product safety. The BfR conducts its own research on topics that are closely linked to its assessment tasks.

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